(a) Real contacts

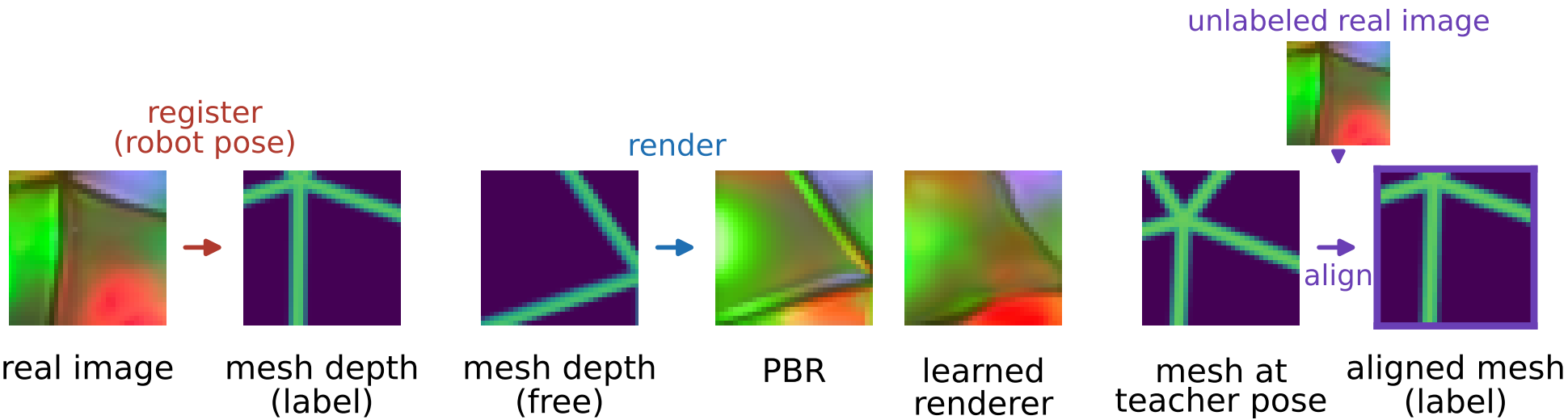
appearance is right;
ch label needs a registered pose

(b) Simulated contacts

labels are free; appearance gap,
renderer fitted on real pairs

(c) AlignTac (ours)

real image for appearance,
aligned mesh geometry for labe



cost: a pose per label | renderer fit on real pairs | K registered contacts, then unlabeled images